

StoryLensEdu: Personalized Learning Report Generation through Narrative-Driven Multi-Agent Systems

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ABSTRACT

Personalized feedback plays an important role in self-regulated learning (SRL), helping students track progress and refine their strategies. However, current common solutions, such as text-based reports or learning analytics dashboards, often suffer from poor interpretability, monotonous presentation, and limited explainability. To overcome these challenges, we present StoryLensEdu, a narrative-driven multi-agent system that automatically generates intuitive, engaging, and interactive learning reports. StoryLensEdu integrates three agents: a Data Analyst that extracts data insights based on a learning objective-centered structure, a Teacher that ensures educational relevance and offers actionable suggestions, and a Storyteller that organizes these insights using the Hero’s Journey narrative framework. StoryLensEdu supports post-generation interactive question answering to improve explainability and user engagement. We conducted a formative study in a real high school and iteratively developed StoryLensEdu in collaboration with an E-learning team to inform our design. Evaluation with real users shows that StoryLensEdu enhances engagement and promotes a deeper understanding of the learning process.

Index Terms: Human-centered computing—Visualization

1 INTRODUCTION

Self-regulated learning (SRL) is widely recognized as one of the most effective learning strategies, empowering students to take charge of their own learning by setting goals, monitoring progress, and reflecting on outcomes [3,47]. Central to this process is personal feedback [4], which provides essential insights that help learners adjust their strategies and improve performance [52]. To support SRL, many online platforms such as Leetcode¹ and Quizizz² offer personalized feedback based on learners’ practice and exam histories, helping them track progress and refine their approaches.

Given the pivotal role of feedback, recent research has increasingly focused on how to generate more effective student reports. The first direction is to enrich the content of these reports. Moving beyond traditional formats that only display grades or raw scores, modern reports now incorporate detailed analytics, such as practice frequency, time-on-task, accuracy [50], and mastery of learning objectives [64], to provide a deeper understanding of student performance. Though more insightful, the growing complexity of report content has outpaced the capabilities of traditional text-based reports. To address this, many systems now integrate visualizations to enable more engaging communication. For example, learning analytics (LA) dashboards [17,38] is a commonly used format that organize student data into interactive, structured visuals. Susnjak *et al.* [50],

for instance, visualize weekly student activity to support exploration of learning patterns.

However, despite their strengths, existing solutions, especially LA dashboards or text-based reports face notable limitations. First, the sheer volume and complexity of the data can overwhelm students, making it difficult to interpret insights without proper guidance. Misinterpretations can also lead to misguided learning decisions [17,38]. Second, dashboards often rely heavily on monotonous presentations (*i.e.*, direct list of charts and text), which may lack engagement, particularly for younger audiences like middle and high school students. This can result in overlooked opportunities for reflection and growth. Third, these reports are typically one-way, requiring students to passively receive and interpret the feedback without the ability to ask follow-up questions for further explanation or guidance.

This research aims at a system for providing personalized feedback that is comprehensive, comprehensible, and engaging, and for providing question-answering assistants to personalize the feedback consumption process. To inform the system design, we first conducted a formative study in a practical high school setting to explore current practices and preferences around personalized feedback from both student and teacher perspectives. We derived a set of design considerations about the generation workflow, content requirements, and presentation format for the report. Based on the findings, we developed StoryLensEdu, a narrative-driven multi-agent system that automatically generates personalized, engaging, and interactive learning reports. The system was iteratively developed in collaboration with an actual E-learning platform team, including two key modules: The first one is a *report generation engine* with a multi-agent architecture. The engine features a team of three LLM-powered agents with different roles. The *Data Analyst* extracts key insights from student data based on a learning objective-centered structure. The generated insights are next evaluated by the *Teacher* agent to ensure its educational value and offer customized suggestions. Then, the *Storyteller* organizes these insights and suggestions into a cohesive and engaging narrative with human-like storytelling techniques. Specifically, we embed the Hero’s Journey narrative framework [14,56] into the Storyteller agent. The design allows the storytelling agent to transform the report into a compelling story, enriched with text and visuals that mirror the learner’s journey across different stages. To further support personalized and interactive exploration, we carefully designed an *interaction module*. It enables post-generation question answering based on user-selected visual or textual elements within the report. Finally, we evaluate StoryLensEdu with real-world users and demonstrate its effectiveness in enhancing engagement and delivering actionable insights.

The main contribution is a multi-agent system architecture that automates context-aware visualization recommendation by dynamically coupling data insights with narrative-driven storytelling. The seamless coupling of automated insight extraction, narrative framing, and interactive Q&A represents a meaningful advancement in how visualization systems can scaffold complex data for non-expert users.

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¹<https://leetcode.com/>

²<https://quizizz.com/>

2 RELATED WORK

2.1 Student Performance Feedback

While traditional text-based reports remain foundational for delivering summative and formative feedback to students, their dense textual format often results in cognitive overload, limiting their effectiveness in facilitating students' reflection [16].

To address these limitations, Learning Analytics (LA) Dashboards have emerged as a promising solution. Susnjak et al. [50] developed an innovative dashboard employing machine learning algorithms to provide predictive insights, enabling students to interpret their learning progress and make informed adjustments. Additionally, Yan et al. [61] demonstrated that integrating GenAI into LA dashboards enables more effective analysis of textual data, unlocking deeper insights into learning processes.

Although LA dashboards mitigate textual overload through visual encoding, they are hard to interpret and fail to sustain engagement. Pozdniakov et al. [38] found that teachers can benefit a lot from explicit data storytelling guidance when interpreting data dashboards. Fernandez-Nieto et al. [17] demonstrated how automated feedback can be transformed into data stories tailored to teachers' instructional goals, fostering reflective student practices. Some researchers also explore alternatives such as visualization, storytelling, and narrative structures to move beyond traditional dashboard paradigms [13, 16]. Ruan et al. [39] showed that embedding analytics within narratives significantly enhanced children's engagement and conceptual understanding. Furthermore, Milesi et al. evaluated the effectiveness of data comics in helping students visualize and reflect on personalized learning experiences [33]. However, these narrative-driven reports are still constrained by a lack of interactive features [6, 49] and fail to deliver a bidirectional analytical experience that enables users to delve deeper into the insights.

Existing feedback formats generally follow two paradigms: static summarization (text-based reports) and exploratory visualization (LA dashboards). While reports often lead to passive reception, dashboards require high data literacy to bridge the gap in interpretation. Our work introduces another paradigm, interactive automated narrative synthesis, combining automated synthesis with post-generation Q&A. Unlike dashboards that prioritize data accessibility for manual exploration, our system prioritizes interpretative scaffolding. By shifting the burden of synthesis from the student to a multi-agent system, we transform fragmented metrics into a cohesive learning journey, moving beyond what the data is to what it means for the learner's progress.

2.2 Data-Driven Article Generation

Data-driven articles are a popular form of data storytelling that combine text, visualizations, and their interplay [22]. To streamline the traditionally manual process of writing narratives and crafting corresponding visualizations, researchers have proposed various interaction paradigms and automation techniques [27]. One key direction supports tighter text-visual integration, through interactive manipulation [26], real-time annotation [32], markup languages [10, 11], programming interfaces [24], and block-based editing [5, 7]. With the rise of large language models (LLMs), recent work increasingly focuses on automating article generation [28]. For example, DataTales [48] uses LLMs to generate textual narratives from charts; SNIL [8] produces episode-based sports news from user input with support for post-editing; and DATAWEAVER [18] enables bidirectional composition between text and visualizations.

While these systems have significantly improved the learnability of data article creation, they often place less emphasis on enhancing expressiveness, especially in domain-specific contexts like education. In contrast, our work leverages LLMs to automate the data article generation pipeline while embedding narrative structures (e.g., the hero's journey [14, 34, 56]) to boost learner engagement and expand the expressive application in educational scenarios.

2.3 Multi-Agent Systems for Data Storytelling

Multi-agent systems have been increasingly adopted across various domains for their ability to coordinate specialized agents toward complex, goal-oriented tasks [30, 54, 58]. These systems typically decompose workflows into modular roles, each handled by an autonomous agent with domain-specific capabilities. Such paradigms have been shown to be more effective than single-agent approaches in handling complex scenarios that involve multiple subtasks [20]. For example, Data Director [42] integrates agents representing a data perceiver, analyst, and designer to automatically generate data videos from tabular inputs. InsightLens [57] employs LLM-based agents to capture, organize, and visualize insights throughout the analysis process, supporting insight navigation through multi-faceted conversational contexts. LightVA [63] introduces a planner-executor-controller loop for task planning and execution in visual analytics. LAVE [53] leverages LLM-powered agents to assist in video editing through language-based augmentation.

Building on this line of work, our system adopts a multi-agent architecture composed of a data analyst, storyteller, and teacher agent, designed to collaboratively generate personalized educational learning reports. This design reduces the risk of conflating these responsibilities and makes the trade-offs between data correctness, educational intent, and narrative quality more manageable.

3 LEARNING OBJECTIVE-CENTERED DATA STRUCTURE

At the core of the system is a learning objective-centered data structure, modeled as a directed graph of interconnected learning objectives. This structure systematically tracks performance, aligns feedback with curriculum standards, and enables downstream agents to coherently extract, reason, and present learning insights. By explicitly linking a learner's knowledge state to defined objectives, the system ensures that feedback is directly referenced to transparent criteria, aligning closely with principles that effective feedback should be anchored to explicit goals rather than provided in isolation [35].

Learning Objective. A learning objective is a specific knowledge point that students should learn during an instructional unit. We define a set of learning objectives that need to be mastered by a specific group of students as: $O = \{obj_1, obj_2, \dots, obj_n\}$. These learning objectives may exhibit dependencies among them, which can be represented as a directed graph $G = (V, E)$, where V is the set of vertices, representing the learning objectives O , and E is the set of directed edges, indicating the dependencies between the learning objectives. For any two learning objectives obj_i and obj_j in O , if there exists a directed edge from obj_i to obj_j , which is denoted as $(obj_i, obj_j) \in E$, then obj_i is referred to as a predecessor objective, indicating that obj_i serves as a prerequisite for mastering obj_j . The graph is automatically generated by parsing official curriculum standards. The system maps specific knowledge points as nodes and their predefined prerequisite relationships as directed edges.

Based on the learning objective graph and the structural components of learning objectives within a student practice question, we define three types of learning objectives (Fig. 1): **(1) Current learning objective:** The learning objective that the user or system is currently focusing on; **(2) Predecessor learning objectives:** The prerequisites for the current learning objective; **(3) Associated learning objectives:** Other learning objectives are always considered alongside the current learning objective.

Student errors are often interdependent. The three types capture these dependencies from distinct perspectives, enabling meaningful feedback and personalized review [31, 59].

Attributes of Learning Objectives. Each student can be modeled as an instance of an abstract learning objective graph, where each node corresponds to a learning objective enriched with a set of attributes (e.g., volume of exercises, time spent, accuracy, *etc.*). The overall learning behaviors and outcomes of a student s on a specific learning objective obj_j are represented as a multi-dimensional time series

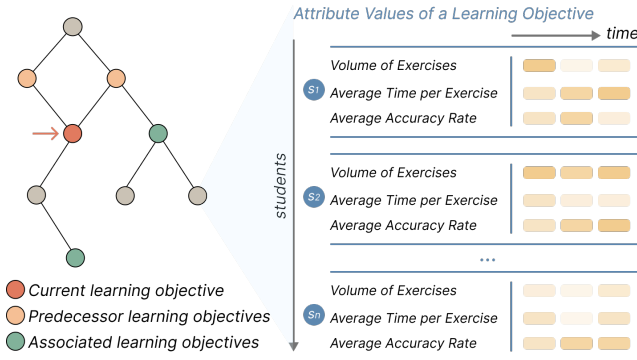


Figure 1: Learning objective graph with three types of objectives.

L_j , capturing the temporal dynamics of these attributes. Each node in the graph also encodes data from multiple students, enabling peer-based performance comparison.

We next describe how to extract L_j for each student based on our example dataset in a real-world E-learning platform. While we focus on a core set of attributes in this paper, the approach is extensible to incorporate additional dimensions of learning behavior in future applications.

First, let S denote the set of a specific student group, and for each student $s \in S$, we define a question set $Q_s = \{q_1, q_2, q_3, \dots, q_i\}$, where q_i represents the i -th exercise attempted by the student. Each exercise q_i is associated with a set of attributes A_i to characterize it. In our case, A_i can be described as $A_i = \{t_i, d_i, u_i, o_i, c_i\}$, where t_i represents the absolute timestamp of when the exercise q_i was initiated by the student s ; d_i represents the time duration taken to complete the exercise q_i ; u_i indicates whether the exercise was completed correctly; $o_i \subseteq O$ denotes a set of learning objectives associated with the exercise q_i ; c_i denotes the difficulty (challenge degree) of the question, i.e., easy, medium, hard.

Next, we can extract the values of the three attributes in a time interval T_k as follows:

- **Volume of Exercises:** The number of exercises $N_j(T_k)$ attempted by the student s related to the objective obj_j during the time interval T_k .
- **Average Time per Exercise:** The average time taken by student s to complete exercises with obj_j during the time period T_k can be calculated as:

$$\bar{d}_j(T_k) = \frac{\sum_{i \in I_j(T_k)} d_i}{N_j(T_k)}$$

where $I_j(T_k)$ is the set of exercises related to obj_j attempted by student s during the time period T_k .

- **Average Accuracy Rate:** The average correctness $\bar{u}_j(T_k)$ for exercises associated with obj_j during the time period T_k , calculated as:

$$\bar{u}_j(T_k) = \frac{\sum_{i \in I_j(T_k)} u_i}{N_j(T_k)}$$

where u_i indicates the outcome of each exercise (1 for correct, 0 for incorrect).

Thus, the overall learning behaviors and outcomes of student s on a learning objective obj_j over time can be represented as a sequence:

$$L_j(s) = \{(N_j(T_k), \bar{d}_j(T_k), \bar{u}_j(T_k))\}_{k=1}^K$$

Given a student s , this sequence encapsulates the evolution of learning behaviors and outcomes with a specific learning objective obj_j across the specified time intervals.

4 STORYLENSEDU

Based on the learning objective-centered data structure and the formative study insights (Please refer to supplementary materials for more details), we developed a multi-agent system, StoryLensEdu, for report generation.

4.1 Overview

This modular design separates complex educational tasks, such as data analysis, instructional reasoning, and narrative construction, into specialized roles. The system overview is illustrated in Fig. 2. A student's self-regulated learning process continuously generates personal data (a), including exercise records, time logs, and performance across learning objectives, based on the learning objective graph (b). Within the automatic *Report Generation Engine* (c-e), the Data Analyst Agent (c) retrieves and structures student learning records based on student ID and learning unit. Leveraging the learning objective-centered graph, it extracts multi-dimensional performance sequences for each relevant learning objective, then mines key insights from these sequences across several dimensions (e.g., trend, anomaly, comparison). It also visualizes the insights to support interpretability. These insights are then sent to the Teacher Agent (d), which contextualizes them within instructional goals. It selects and prioritizes feedback based on both the student's current performance and learning objective dependencies (e.g., prerequisites and associations), and crafts pedagogically meaningful suggestions. Next, the Storyteller Agent (e) integrates both the analytical insights and instructional feedback into a coherent narrative using the Hero's Journey structure. It assembles story elements aligned with the student's progression and refines the tone and content to ensure the report is both informative and engaging. The output is a personalized learning report (f), combining narrative text, performance insights, and rich visualizations. Students can further interactively explore the report via the *Interaction Module* (g). They can select report segments or visual elements and ask questions, and the system will generate multimodal responses grounded in both the selected context and the underlying learning objective graph (h).

From an educational theory perspective, this unified, objective-driven workflow addresses all three key questions of effective feedback identified by Hattie and Timperley [23], i.e., “Where am I going?” (the learning objectives define the goals), “How am I going?” (the Data Analyst and Teacher agents identify the student's current performance status), and “Where to next?” (the Teacher and Storyteller agents provide suggestions on next steps). It operationalizes formative assessment by not only diagnosing learning gaps but also immediately offering adaptive feedback [2].

4.2 Data Analyst Agent

The Data Analyst is primarily responsible for extracting insights from our organized student learning data to provide summative feedback and formative feedback.

4.2.1 Summative Feedback

For summative feedback, we employ an automatic approach to mine insights from the constructed multivariate sequences L_j . In our data formulation, Assessment Mode (Exercise or Test) and Learning Objective are treated as categorical dimensions alongside the temporal sequences. Consequently, the input to our algorithm is a multi-dimensional dataset where each record contains the performance metrics L_j associated with specific dimension values.

For summative feedback, our system processes these multi-dimensional datasets to identify significant learning patterns. First, the mining algorithm iterates through the subspaces defined by these dimensions. This allows us to extract data insights across diverse insight types (e.g., majority, outlier, trend, change point, and low variance) in different dimensions. Then the algorithm outputs raw

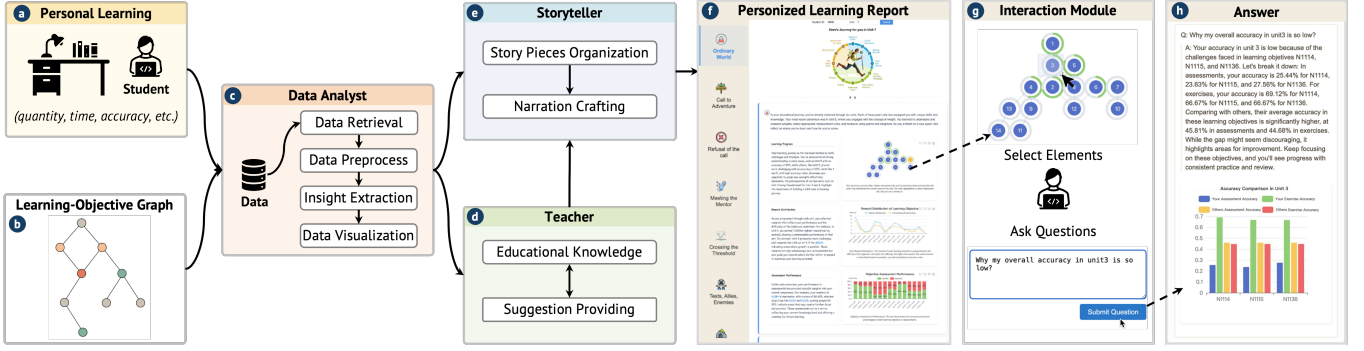


Figure 2: System Overview. The system takes student learning data (a) structured around a learning objective graph (b), which is analyzed by the Data Analyst agent (c) to generate diagnostic insights and visualizations. These are embedded into an annotated objective graph, then interpreted by the Teacher agent (d) to produce pedagogically grounded feedback. The Storyteller agent (e) organizes the content into narrative components using a Hero’s Journey structure and composes the personalized learning report (f). Users can interact with the report (g) to explore details and access contextual explanations (h).

insight tuples containing the insight type (e.g., Outlier), dimension (e.g., specific learning objective, accuracy), and the specific data evidence. These patterns offer students multifaceted views of their performance, such as identifying typical behaviors (majority), anomalies (outliers), or significant shifts (change points), which are critical for self-monitoring and reflection.

To prioritize the most informative insights, we adopt the significance scoring framework proposed by Tang *et al.* [51], which ranks insights based on a composite of statistical significance (p-value) and market share analysis [12]. We surface the top three-ranked insights to help students efficiently recognize key performance signals and take informed, targeted actions. Finally, to ensure precision, these findings are algorithmically pre-calculated as a quantitative ground truth rather than being generated by the LLM. Following a metadata-first workflow, the Teacher agent acts as a deterministic tool-user, retrieving this immutable structured JSON layer to generate interpretable natural language explanations (see Fig. 3(G)).

4.2.2 Formative Feedback

For formative feedback, we first identify the learning objectives that need attention and suggestions. Specifically, this agent continuously maps students’ data onto the relevant learning objectives. To estimate the difficulty and instructional value of a learning objective Obj_i , the Data Analyst agent first retrieves all relevant questions tagged with Obj_i from the database. It then calculates the distribution of these questions across difficulty levels (easy, medium, and hard), expressed as proportions of the total. Each difficulty level is assigned a weight, and a weighted sum is computed to derive a scalar *reward score* for Obj_i . This score reflects the potential instructional gain or effort required to master the objective—higher values correspond to more challenging objectives.

After a student completes a learning unit, the system computes an actual reward by combining question difficulty with the student’s accuracy. In parallel, the Data Analyst calculates several additional indicators per objective in the learning graph, such as estimated mastery level (e.g., 85% mastery of Obj_i), frequency of common errors, and learning velocity (*i.e.*, improvement over time). Collectively, these indicators form a structured diagnostic profile that quantifies student progress, aligns with curriculum goals, and supports objective-specific feedback [36].

For each selected learning objective, we analyze it within the broader context of the learning objective graph. Unlike traditional data storytelling approaches [42, 55], which primarily rely on flat or hierarchical data table relationships, our method captures topological and semantic relationships among nodes, structured through the

learning objective graph [29]. These relationships, grounded in the three defined types (current, predecessor, and associated learning objectives), enable more meaningful insights and relevant feedback.

- **Current learning objective.** The learning outcomes of the current learning objective may be closely related to its practice process. So we consider the insights extracted for this node in the summative feedback as potential formative feedback.
- **Predecessor learning objectives.** The reasons for insufficient learning of the current learning objective may be related to the need for improvement in its predecessor nodes. Therefore, we apply the same insight extraction approach from the multi-dimensional time series of its predecessor nodes. Notably, if a predecessor node has further predecessors, we continue exploring backward.
- **Associated learning objectives.** Students may perform well on questions involving a single learning objective but may struggle when multiple learning objectives are associated. To determine whether suggestions should be provided based on associated learning objectives, we organize the multivariate sequences L_j for a associated objective set $J = \{j_1, j_2, j_3, \dots, j_q\}$, where J is summarized from the questions practiced by students. Specifically, we discard associated objective sets that include any predecessor nodes of the current learning objective, as we have already considered them separately.

The extraction method follows that described in Sec. 4.2.1. Additionally, the extracted insights across the three types are also contextualized through peer comparison to highlight relative performance. This processed information (the updated objective graph with performance annotations) is then passed along to the Teacher agent. By operating at the granularity of learning objectives, the Data Analyst agent essentially answers the formative question: “*Where is the learner right now with respect to each learning goal?*” This objective-centric diagnosis is crucial for effective formative feedback [2, 23]. Rather than giving a single overall score or generic remarks, the system can say, e.g., “*you have mastered Objectives A and C, is making progress on B, but is struggling with Objective D.*” Such specificity not only empowers more actionable feedback downstream but also aligns with established principles in educational assessment, which emphasize the value of tying feedback directly to learning goals [35].

4.3 Teacher Agent

The Teacher agent receives the diagnostic performance map from the Data Analyst and creates pedagogically meaningful feedback for each learning objective. It leverages pedagogical rules, domain knowledge, and best practices to determine the appropriate guidance

Table 1: Adaptation of hero’s journey narrative framework to personalized learning report generation.

Hero’s Journey Narrative Framework		Adaptation to Personalized Learning Report Generation		
Phase	Stage	Mappings to Learning Stages	Story Piece Description	Information
Departure	S.1 Ordinary World	Review prior knowledge	Prior learning objectives and performance	Overview and Introduction
	S.2 Call to Adventure	Start a new learning unit	Introduction to this unit’s objectives	
	S.3 Refusal of the call	Anxiety about a new unit	Introduction to the new unit challenges	
Initiation	S.4 Meeting the Mentor	Learning through lectures or guidance	Emphasis on acquired knowledge	Summary Information
	S.5 Crossing the Threshold	Preparing for exercises	Introduction to upcoming unit exercises	
	S.6 Tests, Allies, Enemies	Practicing with various problems	Insights based on exercise performance	
Unification	S.7 Approach to the innermost cave	Preparing for the test	Emphasis on accumulated practice	Formative Guidance
	S.8 The Ordeal	Taking the unit test	Exam performance with comparisons	
	S.9 The Reward	Mastering basic unit knowledge	Summary of what has been learned	
Return	S.10 The road back	Completing unit learning tasks	Learning task completed	Formative Guidance
	S.11 The resurrection	Receiving feedback and taking action	Suggestions based on performance	
	S.12 Return with the Elixir	Ready to move on with full understanding	Hope for future improvement	

for the student. For each objective, the Teacher agent evaluates the student’s performance (mastery level, errors, attempts, *etc.*) and generates tailored feedback. If the student has demonstrated strong mastery of an objective, the Teacher agent will respond with positive reinforcement and perhaps an extension suggestion (e.g., “*You have a solid understanding of Objective A. Great job! You might try a more challenging problem involving this concept to deepen your skills.*”). If the student is struggling with an objective, the Teacher agent will provide constructive, specific advice on how to improve (e.g., “*Objective B is still challenging for you. I noticed confusion in applying formula X – reviewing the underlying principle of Y and practicing those types of problems will help. Let’s focus on that, and remember the strategy we discussed...*”). In cases of partial understanding, the agent might both praise what is correct and coach the student on what is missing, following the “medal and mission” approach (highlighting successes and next steps) [15].

By centering comments on how the student performed relative to the objective’s criteria, the agent ensures the feedback is informative (about the work) and not judgmental (about the person). The Teacher agent essentially acts as an automated tutor, emulating the role of a human teacher engaging in formative assessment. It uses the evidence of learning (from the Data Analyst) to adapt its instruction and guidance to the learner’s needs [2]. Research has shown that such formative feedback, when done well, can substantially boost learning gains [2], especially for students who are struggling [1], by explicitly addressing misunderstandings and providing clear routes for improvement. The design of the Teacher Agent also draws on the principles of cognitive apprenticeship [9], which emphasize modeling expert thinking, providing scaffolded guidance, and fostering learner independence. For example, the agent makes expert thinking visible, explaining not just what is right or wrong, but why (e.g., “Revisiting concept Y will help you address challenges in Objective B”) based on the learning objective graph.

4.4 Storyteller Agent

The Storyteller agent is responsible for synthesizing the various objective-level feedback pieces from the Teacher agent and the insights extracted from the Data Analyst agent into a coherent, personalized feedback report that is clear and engaging for the student. While the Teacher agent focuses on what to say instructionally, the Storyteller agent focuses on how to say it in a narrative form that the learner will find accessible and motivating.

A list of disjointed bullet points per objective can be factual but might not be very motivating or easy to digest. In contrast, a narrative format helps the student see the big picture of their learning journey: how the objectives they have mastered are accomplishments to be proud of, and how the challenges they face are part of a normal learning process with specific remedies available. Moreover, the Storyteller carefully calibrates the feedback tone to maintain the student’s motivation and self-efficacy. It balances praise and

constructive critique so that the student remains confident to improve without feeling discouraged or complacent.

Based on the performance insights and instructional suggestions generated by the Data Analyst and Teacher Agents, the Storyteller Agent employs the Hero’s Journey narrative framework to organize the report into a coherent and engaging story structure. The Hero’s Journey is a twelve-stage narrative archetype that has been widely used across educational and storytelling contexts to enhance emotional engagement and perceived relevance [14, 34, 56]. We selected the Hero’s Journey over simpler structures, such as the three-act model, because its cyclical and transformative nature aligns more closely with the iterative process of self-regulated learning. While a three-act structure (setup, confrontation, resolution) captures linear progress, the Hero’s Journey emphasizes the internal transformation of the protagonist through setbacks and trials. This reframes learning not as a series of scores, but as a journey where “Tests” (S.6) and “Ordeals” (S.8) are essential, non-permanent stages of growth. Furthermore, the 12-stage framework provides the granularity necessary to map specific educational milestones, such as prior-knowledge review (S.1), formative practice (S.6), and post-exam reflection (S.11), to distinct narrative beats, ensuring the feedback is both cohesive and actionable.

Specifically, we map each student’s personalized learning process for a given unit to the twelve stages of the Hero’s Journey (see Tab. 1). Each stage corresponds to a distinct story piece, capturing elements such as initial goals (e.g., mastering learning objectives), encountered difficulties (e.g., exercises), turning points (e.g., exams), and reflections (e.g., teacher’s suggestions). To manage cognitive load, the twelve stages are hierarchically grouped into four macro-phases (Tab. 1), following an “overview-first, detail-on-demand” logic. Stages 1–3 provide a high-level review of goals, stages 4–9 focus on core performance analysis, and stages 10–12 deliver actionable guidance. This structure ensures students grasp the big picture before exploring specific diagnostic details. Furthermore, several stages serve as brief narrative transitions (e.g., S.4, S.10) to maintain story flow without adding analytical density. The result is a report that not only delivers informative feedback but also promotes reflection and engagement through a personalized, story-driven format. An report example is shown in Fig. 3.

4.5 Post-Generation Interaction

To support deeper understanding, we developed an interaction module for follow-up Q&A with the generated report.

4.5.1 Interaction Design

Students can initiate contextual inquiries by selecting specific narrative segments or visual elements within the generated report. The system then generates detailed responses, enriched with visual explanations when necessary [44, 45]. Fig. 4 demonstrates this interactive questioning process. The student uses the visual selection tool to

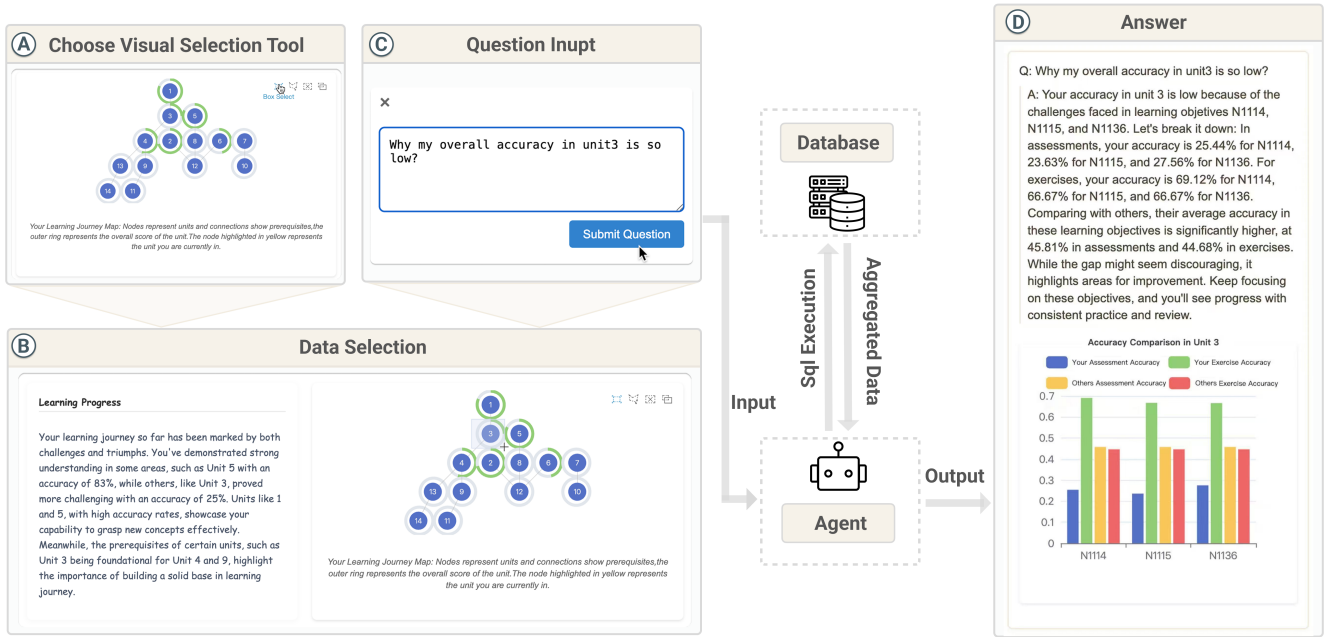


Figure 4: (A) Choose a visual selection tool (e.g., box or lasso). (B) Brush to choose the data points of interest. (C) Type the question and then submit it. (D) The agent retrieves related data from the database to generate the answer, and returns textual and visual elements.

peer averages to clarify potential areas for improvement. Students may also select combined textual and visual elements to generate queries, and the system dynamically determine and present the most suitable visualizations to effectively address their inquiries.

4.5.2 Question Answering Model

The Q&A model is tightly integrated with the learning objective-centered data structure and its underlying knowledge graph to support precise, context-aware responses. When a student selects a text or visual element to pose a query, the system first uses the graph's topology to map the selection to relevant learning objectives.

Instead of executing complex, pre-defined analytical queries, the system then performs targeted SQL queries to retrieve a holistic dataset pertinent to these objectives. For single-objective queries, this dataset includes performance insights along three structured dimensions: the current objective, its predecessors, and its associated objectives. This tri-level analysis helps uncover both surface-level and latent learning difficulties. For multi-objective queries, data is retrieved across all selected objectives to identify higher-order patterns in interrelated learning behaviors. To enrich interpretability, peer comparison metrics are also incorporated into the dataset.

This curated dataset, combined with the student's natural language query and a predefined prompt template, is dynamically engineered into a detailed prompt. The LLM then processes this prompt, performing an in-context analysis to formulate a natural language answer while simultaneously generating the Echarts configuration for an appropriate visualization. This LLM-centric architecture bypasses the rigidity of traditional query systems, allowing for a more intuitive and responsive analytical dialogue.

We utilize an asynchronous pipeline where student metrics are pre-calculated and cached. This separates heavy data aggregation from interaction, ensuring that real-time report generation is limited only by LLM inference time (5s) rather than database queries.

5 USAGE SCENARIO

In this section, we will use fictional data to go through the report generated automatically by StoryLensEdu. Assume a student named

Steven, who just finished studying Unit 7 of the third-grade curriculum. After completing all the practice and exam questions, he hoped to see his learning progress. Therefore, he generated a report with StoryLensEdu, as shown in Fig. 3.

When Steven first started reading the report, which said, *"In this journey, you've already ventured ..."*, he felt as if he had entered an adventure world. Through the learning journey map (Fig. 3-A), he realized that he had mastered many units well, such as Unit 1, Unit 2, and Unit 5, which made him feel very proud. However, he also acknowledged some shortcomings, such as in Unit 3, where the report showed he might have only scored 25%. Through more detailed line charts and bar charts (Fig. 3-B and C), he learned that concepts N1114 and N1115 in Unit 3 were areas he had not mastered well, scoring around 24% to 25%. Steven silently resolved to solidify his understanding of Unit 3.

After that, Steven started to carefully examine the assessment related to Unit 7. Through the reward distribution visualization (Fig. 3-D), he discovered that Unit 7 had four concepts he needed to master. Among them, S1205 and S2106 appeared to be the most challenging. He also saw the difficulty distribution (Fig. 3-E). For example, about 49.02% of the questions related to S1206 were rated as difficult. This allowed him to recognize how challenging this unit is. At the same time, the report's hints reminded Steven of those days when he faced tough questions every day. *"Have I really defeated the enemies?"* Upon examining the data, he found a 93.75% mastery rate in S1102 (Fig. 3-F). Each question appeared to take an average of about two minutes. He felt a quiet sense of satisfaction. However, he also noticed that he did not seem to have mastered S1205 well, which would require attention in the future.

He continued to read, *"If those practice questions were the enemies, then what I face next could be the boss."* When he examined the exam-related assessments (Fig. 3-G), the accuracy rates for all four learning objectives exceeded 80%, indicating solid overall mastery. The report also showed him various information regarding his time spent on questions, and its encouraging tone helped sustain his passion. *"I still need to improve."* He noted some insights mentioned further down the report, such as not having enough difficult questions in S1102. The report mentioned that Steven had done

quite well in mastering the concepts of Unit 7. *“There is no reward more important than this recognition,”* Steven stated.

Finally, it was essential to focus on reflection and feedback. *“The report showed me ways to continue improving, such as suggestions for S1206.”* Through this report, Steven felt as if he had re-experienced his learning journey in Unit 7. He could see himself at the beginning, feeling lost and afraid of difficulties, then pushing forward, overcoming challenges... This immersive report guided him through his experiences in Unit 7 and helped him digest and absorb essential learning summaries and suggestions.

6 USER STUDY

Our evaluation aimed to assess the intuitiveness, engagement, and interactivity of the automatically generated personalized learning reports produced by StoryLensEdu.

6.1 Participants and Procedure

We invited 10 students (S1 - S10) and 4 teachers (T1 - T4) from a high school. Each student received their personalized report automatically generated by StoryLensEdu based on real-world data, which includes their own learning data over two semesters and relevant data from over 1,000 peers. Students were asked to read their reports carefully, employing a think-aloud method to articulate their thought processes. During their reading, students were asked to freely interact with the Q&A module, exploring further details and raising inquiries as needed. Teachers randomly explored all students' reports. Finally, we carried out individual semi-structured interviews to gain deeper insights into their experiences and asked each participant to complete a questionnaire. The protocol has been approved by the IRB in our institution. All learning data is de-identified before transmission to LLM APIs. Personal identifiers (e.g., names, student IDs) are replaced with anonymized tokens.

6.2 Quantitative Results

The results from the questionnaire, as shown in Fig. 5, indicated generally positive feedback among both students and teachers. Specifically, clarity of data insights (mean=4.5, SD=0.53 for students; mean=4.5, SD=0.58 for teachers), intuitiveness of charts (mean=4.6, SD=0.52 for students; mean=5.0, SD=0.00 for teachers), and interactive exploration (mean=4.6, SD=0.52 for students; mean=4.8, SD=0.50 for teachers) were particularly praised, underscoring the system's effectiveness in visualizing complex educational data intuitively and interactively. However, relatively lower ratings were observed in narrative engagement (students mean=3.4, SD=0.52; teachers mean=3.5, SD=0.58) and the connection of storytelling with personal learning behaviors (students mean=3.3, SD=0.48; teachers mean=3.3, SD=0.50). This indicates an opportunity to improve the narrative design to more closely resonate with the diverse expectations of the learners.

6.3 Qualitative Findings

Semi-structured interviews revealed rich insights into how participants perceived and interacted with StoryLensEdu. A key theme among students was the sense of agency enabled by interactive elements. All students reported that features such as embedded Q&A modules and annotated visualizations not only clarified their performance gaps but also empowered them to take action. As S4 noted, *“The interactive visuals made it easy to pinpoint exactly where I struggled, and asking questions directly through the interface was really empowering.”* Students consistently described interactivity as pedagogically meaningful, supporting self-reflection and deeper engagement. Teachers similarly valued visuals, particularly for diagnostic purposes. T2 remarked, *“Charts simplified complex data clearly, allowing us to quickly grasp students' progress and intervene when necessary.”* Two teachers (T3, T4) highlighted that visualizations facilitated early intervention and cohort-level comparison.

The narrative component was broadly appreciated for structuring feedback into a coherent and accessible format. Participants described it as reducing intimidation and enhancing interpretability. T4 commented, *“The narrative helps frame feedback in a way that feels more like a conversation than a judgment.”* S9 noted, *“The report is enjoyable and relatable. I could better connect my learning actions with the outcomes shown.”* Many (10/14) endorsed the Hero's Journey structure as logically coherent and pedagogically effective in connecting past performance, current outcomes, and future goals.

In addition, several participants proposed enhancements to improve personal resonance and emotional alignment. S7 suggested adapting the narrative tone and framing to better reflect individual learning preferences. While most students favored supportive language, a few preferred more provocative tones to enhance motivation. Thus, a uniform narrative style may limit engagement, and adaptive emotional framing could enhance relevance. Teachers echoed the need for customization but cautioned against sacrificing specificity for fluency. As T1 noted, *“The narrative sounds nice, but sometimes it glosses over the harder truths. We still need space for precise, critical feedback.”* Participants (S1, S6, S7, T1) also raised concerns about information density, reporting difficulty in locating high-priority content. Suggested improvements included search, filtering, and visual anchors. T1 further proposed using animation to enhance immersion and data interpretability. Others (S2, S10) recommended incorporating imagery or comic-style illustrations to strengthen engagement. Finally, T3 emphasized the need for narrative formats tailored to diverse educational levels and subjects, noting that, *“The storytelling format suitable for senior-level math may not resonate with a lower-grade language class.”*

Failure Case Analysis. While StoryLensEdu scored high in clarity, ratings for narrative engagement (Q5) and connection (Q6) were moderate (3.3–3.5). Qualitative interviews suggest this is not a rejection of the narrative paradigm, but rather a reflection of the “one-size-fits-all” nature of the current Hero's Journey framework. First, we observed narrative redundancy in reports for students with sparse activity data; because the 12-stage framework is currently rigid, the storyteller occasionally repeated the same insights to fill all sections. Second, there was a mismatch in tone, where the “Heroic” persona remained overly encouraging even when a student's accuracy dropped significantly (e.g., calling a student a “brave adventurer” despite a 20% success rate), which can gloss over critical gaps. Third, a prioritization gap occurred between statistical and pedagogical value; for instance, the system might highlight a rapid misclasp as an insight into speed, which is educationally irrelevant. Fourth, the system sometimes exhibits logic leaps in pedagogical reasoning by misinterpreting conceptual associations as causal prerequisites. For example, finding a topological link between “Functions” and “Geometry” might lead the system to conclude that a student's struggle in “Functions” is due to poor “Geometry” skills, while ignoring the actual learning obstacle. Finally, granularity issues in the learning objective graph can lead to vague advice, such as suggesting a general topic review instead of a specific procedural correction.

7 DISCUSSION

Multi-Agent Collaboration. Our system introduces a multi-agent architecture where specialized agents emulate distinct education functions, mirroring human educator roles. This decomposition brings three key advantages. First, it enables functional specialization: each agent can be optimized for a narrow communicative goal, improving clarity and coherence. Second, it supports extensibility: new agents (e.g., peer tutor, parent mediator, or self-reflection companion) can be integrated without overhauling existing components. Third, it opens pathways for explainability. Users can trace the origin of each narrative element to a responsible agent and its rationale. This architecture also introduces opportunities for intra-agent

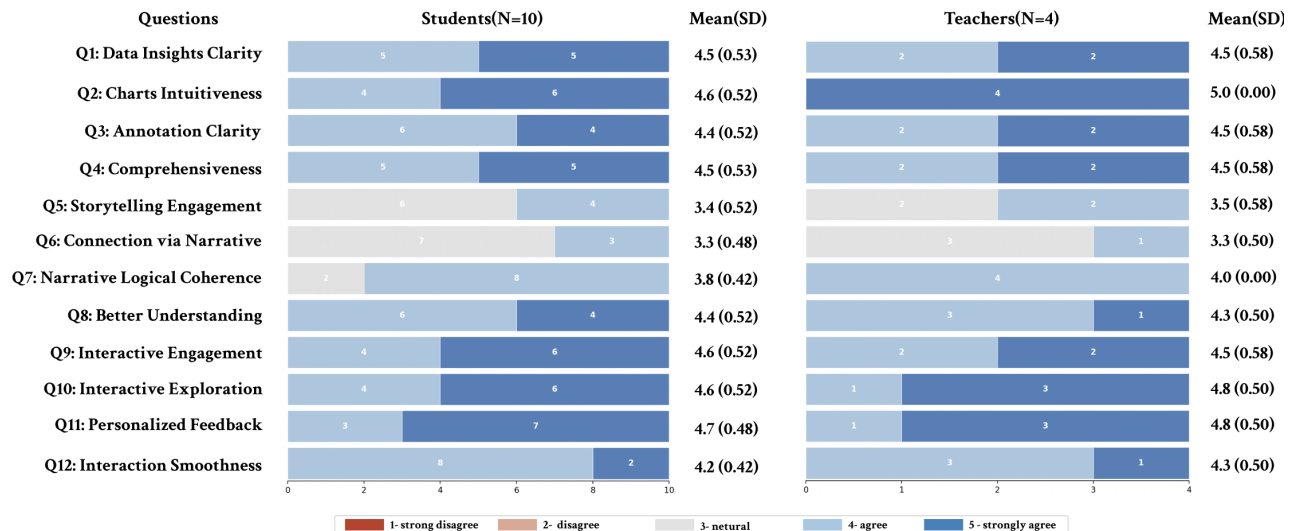


Figure 5: Questionnaire results in the user study for 10 students (S1 - S10) and 4 teachers (T1 - T4).

dialogue and negotiation. For instance, a tutor agent may prioritize conceptual feedback, while a motivational agent might advocate for positive framing. Designing protocols for agent-level conflict resolution or consensus-building could help balance feedback tone and focus in a pedagogically grounded manner [58]. Additionally, agent collaboration may support adaptive feedback pacing: agents could coordinate to stagger guidance over time, fostering sustained engagement rather than information overload.

Personalization. This work demonstrates how student learning reports can be automatically generated with engaging narratives grounded in performance data. Moving forward, personalization can be deepened along several dimensions. One direction is emotional and motivational adaptation: students exhibit varied preferences for narrative tone; some favor encouraging language, while others respond better to more provocative feedback. Future systems could leverage sentiment analysis or learner profiling to dynamically adjust tone and emotional framing [25, 60]. Participants also emphasized the importance of learner diversity across educational levels and domains. The structure, language, and visual density of feedback may require adaptation for different age groups (e.g., primary vs. college) and disciplines (e.g., mathematics vs. programming). Such contextual sensitivity can enhance clarity, relevance, and cognitive accessibility. Personalization should also account for the involvement of additional stakeholders, such as parents or advisors, to support a more holistic feedback ecosystem [21]. This could be enabled through multi-view reports or shared progress dashboards. Personalization can evolve temporally as students’ goals, behaviors, and competencies change. This calls for longitudinal learner modeling and feedback strategies that evolve alongside the learner. Additionally, a current limitation is the heavy reliance on numeric logs, which might overlook specific procedural errors. However, StoryLensEdu’s multi-agent architecture is well-positioned to integrate LLM-based analysis of non-numeric artifacts. Future work could leverage a User Profiler agent to identify student preferences or learning styles. By feeding students’ verbal explanations, LaTeX-formatted derivations, or even OCR-processed sketches into the Data Analyst agent, the system could perform qualitative error pattern recognition. Future iterations will explore linking specific distractor analysis (common wrong answers) to the Storyteller agent, enabling even more nuanced narratives like, “*You consistently confused the sine and cosine rules, a common hurdle that we can overcome by...*”

Generalizability. While evaluated in high school math, which is a prototypical testbed for validating the learning objective graph due to its explicit prerequisites, StoryLensEdu is inherently domain-

agnostic. Adapting to other STEM subjects requires only updating metadata and pedagogical rules, while for less hierarchical domains like the Humanities, the system can pivot from mastery-based to theme-based narrative logic. The system’s modular feedback pipeline, composed of objective-linked performance analysis, narrative construction, and role-specific agent contributions, can scale to contexts such as higher education (e.g., lab reports) and workplace training (e.g., compliance), and can also be extended to support other narrative formats [19, 37, 40, 46, 62]. Such flexibility stems from the structural encoding of objectives and the use of interpretable intermediate representations (e.g., knowledge graphs, reflection arcs), enabling customization without altering core logic. However, effective generalization requires retuning narrative style, granularity, and agent voice to align with the specific cognitive and emotional norms of each target domain [41, 43].

Limitations. The system also presents several limitations. First, it relies on structured, objective-aligned performance data, which may not generalize to domains with unstructured or open-ended tasks. Extending the approach to such contexts will require more sophisticated data interpretation and knowledge extraction methods. Second, our study prioritized evaluating the user experience of the proposed narrative paradigm. Future controlled trials will be conducted to quantify its impact on learning outcomes relative to established baselines, such as traditional dashboards and text summarization. Finally, the system does not support direct teacher input during feedback generation, limiting opportunities for co-authorship. Enabling educator intervention could improve trust, contextual relevance, and classroom integration.

8 CONCLUSION

We presented *StoryLensEdu*, a narrative-driven, multi-agent system for generating personalized learning reports that are structured, intuitive, engaging, and interactive. Beyond automation, we argue that educational feedback should not only inform, but also motivate and guide. To that end, this work opens promising directions for collaborative human-AI feedback systems that evolve with learners, integrate diverse stakeholder perspectives, and adapt across domains and learning contexts.

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